

ThingSpeak Cheat Sheet

@Appleton Innovations

Thingspeak

An IoT analytics platform service that allows you to aggregate, visualize and analyze live data streams in the cloud. It stores and retrieves data from things using the HTTP and MQTT protocol over the Internet.

Features

- Easily configure devices to send data to ThingSpeak using popular IoT protocols.
- Visualize sensor data in real-time.
- Aggregate data on-demand from third-party sources.
- Use the power of MATLAB to make sense of our IoT data.
- Run IoT analytics automatically based on schedules or events.
- Prototype and build IoT systems without setting up servers or developing web software.
- Automatically act on the data and communicate using third-party services like Twilio or Twitter.

Channel

With ThingSpeak, the data is stored in channels. Each channel includes 8 fields to store any type of data, plus three fields for location data, and one for status data. We can create as many channels as we need for our application.

API Reference

REST API:

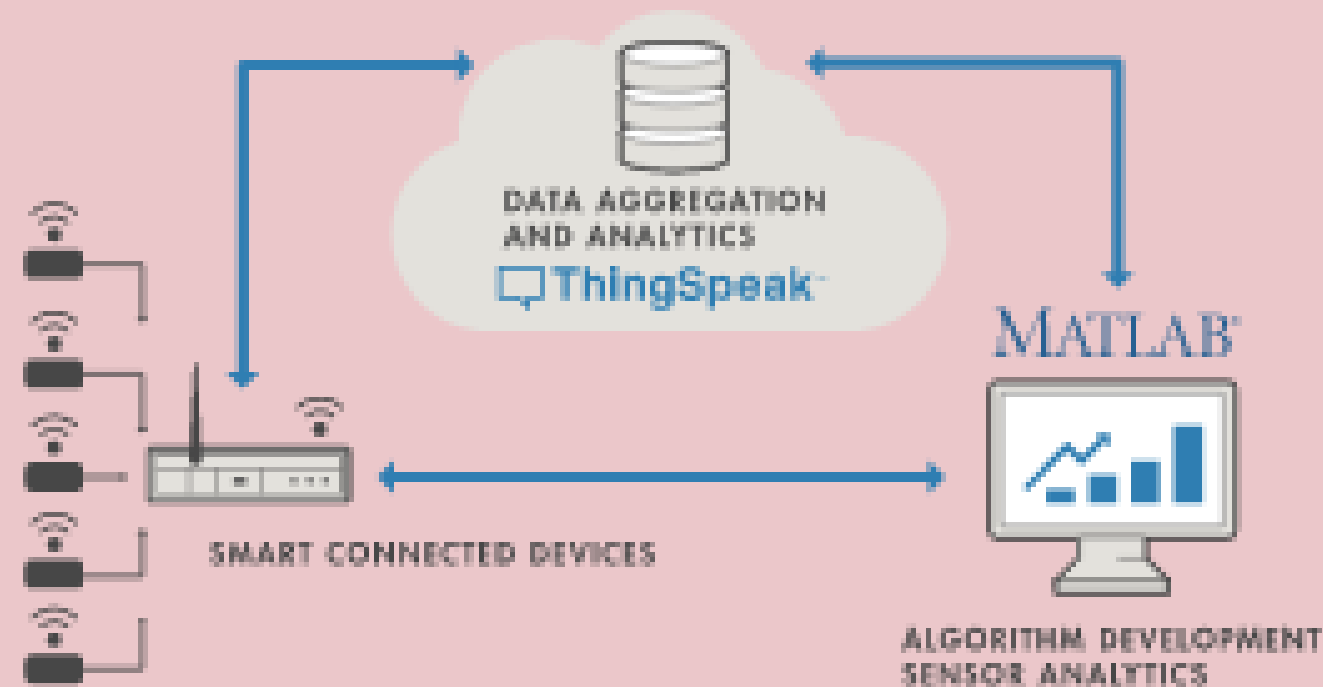
Use REST (Representational state transfer) API calls to create and update ThingSpeak channels and charts. It is designed as a request-response model that communicates over HTTP.

MQTT API:

ThingSpeak enables clients to update and receive updates from channel feeds via the ThingSpeak MQTT broker. MQTT is a publish/subscribe communication protocol that uses TCP/IP sockets or WebSockets.

API (Application Programming Interface): API keys are the passwords to access a channel.

1. To update channel / data logging : API Write Key will be used to access in this mode.
2. To retrieve data : API Read Key will be used to access in this mode.



ThingSpeak Apps

- **MATLAB analysis:** Used to prepare, filter, and analyze data .
- **MATLAB Visualizations:** Visualize data in MATLAB plots like area plot, line plot, stem plot, scatter plot, etc.
- **Plugins:** Plugins enable you to use HTML, CSS, and JavaScript to display custom visualizations and controls on the private view of a channel.
- **ThingTweet:** This allows you to post messages to Twitter via ThingSpeak.
- **ThingHTTP:** This allows you to connect to web services and supports GET, PUT, POST, and DELETE methods of HTTP.
- **TweetControl:** Monitors Twitter feeds for a specific keyword and then processes the request. Once that keyword is found, we use ThingHTTP to connect to a different web service or execute a specific action.
- **React:** Send a tweet or trigger a ThingHTTP request when the Channel meets a certain condition.
- **TalkBack:** Use this app to queue up commands and then allow a device to act upon these queued commands.
- **TimeControl:** Using this app, we can do a ThingTweet, ThingHTTP, or a TalkBack at a specified time in the future.

Functions

thingSpeakWrite : Write data to ThingSpeak channel.

Syntax and description :

thingSpeakWrite(channelID,data,'WriteKey','channel write API key') - writes the data to the specified channel.

thingSpeakWrite(____,Name,Value) - uses additional options specified by one or more Name,Value pair arguments.

response = thingSpeakWrite(____) - returns the response provided by the ThingSpeak server on successful completion of the write operation.

thingSpeakRead : Read data stored in ThingSpeak channel

Syntax and description :

data = thingSpeakRead(channelID) - reads the most recent data from all fields of the specified public channel and returns the data as a numeric type.

data = thingSpeakRead(channelID,Name=Value) - uses additional options specified by one or more Name=Value pair arguments.

data = thingSpeakRead(____,ReadKey='channel Read API key') - uses the ThingSpeak Read API key to read from a private channel.

[data,timestamps] = thingSpeakRead(____) - returns timestamps from the specified channel on and can include any of the input arguments in previous syntaxes.

[data,timestamps,channelInfo] = thingSpeakRead(____) - returns channel information.

API Requests

Write Data : Update channel data with HTTP GET or POST

Eg.: GET https://api.thingspeak.com/update?api_key=yourAPIkey&field1=0

Read Data : Read data from all fields in channel with HTTP GET

Eg.: GET https://api.thingspeak.com/channels/266256/feeds.json?api_key=yourAPIkey&results=2

Read Status : Read status field of channel with HTTP GET

Eg.: GET https://api.thingspeak.com/channels/1480680/status.json?api_key=yourAPIkey